

WINDOWS COMPUTER CLEAN-UP AND VIRTUAL MEMORY PRACTICAL

Department: ICT Department

Practical Title: Windows Update, Cache Cleaning, Temporary File Removal and Virtual Memory Configuration

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Computer Name: OPR-8-LODYA

Department/Office: REGISTRATION

Date: 24th JULY 2024

Supervisor: FRANCIS TUMOTI

1. PRACTICAL OBJECTIVES

By the end of this practical exercise, the intern should be able to:

- Check the computer's available disk space.
 - Create a system restore point.
 - remove temporary files.
 - Clear Windows cache files.
 - Empty the Recycle Bin.
 - Clear the DNS cache.
 - Run Windows Disk Cleanup.
 - Check and repair Windows system files.
 - Scan for Windows updates.
 - Configure Windows virtual memory.
 - Restart and test the computer.
 - Record any issues encountered during the exercise.
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2. IMPORTANT SAFETY INSTRUCTIONS

Confirm that the computer has been approved for maintenance by the ICT Department.

Save and close all open documents before starting.

Confirm that the user has backed up important files.

Do not delete files from the user's Documents, Desktop, Downloads or other personal folders unless instructed by the ICT supervisor.

Do not disable antivirus, firewall or security software.

Do not install unapproved applications.

Use an administrator account only when required.

Report any serious errors to the ICT supervisor.

PART A: COMPUTER INFORMATION

Task 1: Record the computer details

Step 1

Press:

Windows key + R

Step 2

Type:

msinfo32

Step 3

Press **Enter**.

Step 4

Record the following information:

Computer Name: OPR-B-L0001A

Windows Version: WINDOWS 11 PRO EDUCATION / VERSION 25H2

System Manufacturer: HP

System Model: HP ProDesk 400 G6 MT

Installed Physical RAM: 4.00 GB

Processor: INTEL (R) Core (TM) i5 - 9500 CPU @ 3.00 GHz

System Type:

64-bit ✓

32-bit

Completion

Computer information recorded successfully.

Issues encountered

N/A

PART B: CHECK AVAILABLE STORAGE

Task 2: Check the computer's disk space

Method 1: Using File Explorer

1. Press:
Windows key + E
2. Select **This PC**.
2. Check the free space under **Local Disk (C:)**.

Record the result

Total disk size: 475 GB

Free space before cleanup: 143 GB

Used space before cleanup: 332 GB

Completion

Available disk space checked and recorded.

Issues encountered

N/A

Task 3: Check disk space using PowerShell

Step 1

Right-click the **Start** button.

Step 2

Select:

Terminal (Admin)

or:

Windows PowerShell (Admin)

Step 3

Run the following command:

```
Get-PSDrive -Name C
```

Expected result

PowerShell should display:

- Used disk space
- Free disk space
- Total disk capacity

Completion

PowerShell opened as administrator.

Disk space command executed successfully.

Disk space information recorded.

Issues encountered

N/A

PART C: CREATE A SYSTEM RESTORE POINT

Task 4: Create a restore point

A restore point allows Windows settings to be restored if a serious problem occurs during maintenance.

Step 1

Press:

Windows key + R

Step 2

Type:

SystemPropertiesProtection

Step 3

Press **Enter**.

Step 4

Select the **C:** drive.

Step 5

Confirm that protection is set to **On**.

Step 6

Click **Create**.

Step 7

Enter the restore point name:

Before ICT Cleanup

Step 8

Click **Create** and wait for completion.

Completion

System Protection was enabled.

Restore point was created successfully.

Restore point name was recorded.

Issues encountered

N/A

PART D: DELETE USER TEMPORARY FILES

Task 5: Clear the current user's temporary files

Step 1

Press:

Windows key + R

Step 2

Type:

%temp%

Step 3

Press **Enter**.

Step 4

Press:

Ctrl + A

to select all files.

Step 5

Press:

Shift + Delete

Step 6

Click **Yes** to confirm.

Important instruction

Some files may be in use and cannot be deleted. Select **Skip** when Windows reports that a file is currently in use.

Completion

%temp% folder opened successfully.

Temporary files selected.

Temporary files deleted.

Files currently in use were skipped.

Issues encountered

N/A

Task 6: Clear user temporary files using PowerShell

Open PowerShell as Administrator and run:

```
Get-ChildItem -Path $env:TEMP -Force -ErrorAction SilentlyContinue | Remove-Item -Recurse -Force -ErrorAction SilentlyContinue
```

Completion

PowerShell command executed.

User temporary files were cleared.

Issues encountered

N/A

PART E: DELETE WINDOWS TEMPORARY FILES

Task 7: Clear the Windows temporary folder

Step 1

Press:

Windows key + R

Step 2

Type:

C:\Windows\Temp

Step 3

Press **Enter**.

Step 4

Approve the administrator prompt if displayed.

Step 5

Press:

Ctrl + A

Step 6

Press:

Shift + Delete

Step 7

Select **Skip** for files that Windows reports are in use.

Completion

Windows temporary folder opened.

Unnecessary Windows temporary files deleted.

Files in use were skipped.

Issues encountered

N/A

Task 8: Clear Windows temporary files using PowerShell

Run:

```
Get-ChildItem -Path "C:\Windows\Temp" -Force -ErrorAction SilentlyContinue |  
Remove-Item -Recurse -Force -ErrorAction SilentlyContinue
```

Completion

Command executed successfully.

Windows temporary files cleared.

Issues encountered

N/A

PART F: EMPTY THE RECYCLE BIN

Task 9: Empty the Recycle Bin manually

Step 1

Locate the **Recycle Bin** on the Desktop.

Step 2

Right-click **Recycle Bin**.

Step 3

Select:

Empty Recycle Bin

Step 4

Click **Yes**.

Completion

Recycle Bin emptied successfully.

Issues encountered

N/A

Task 10: Empty the Recycle Bin using PowerShell

Run:

`Clear-RecycleBin -Force`

Completion

PowerShell command executed.

Recycle Bin cleared.

Issues encountered

N/A

PART G: CLEAR THE DNS CACHE

Task 11: Flush the DNS cache

The DNS cache stores website and server address information. Clearing it may resolve some internet and website connection problems.

Step 1

Open Command Prompt or PowerShell as Administrator.

Step 2

Run:

```
ipconfig /flushdns
```

Expected message

Successfully flushed the DNS Resolver Cache.

Completion

DNS cache command executed.

Success message displayed.

Issues encountered

N/A

PART H: CLEAR MICROSOFT STORE CACHE

Task 12: Reset the Microsoft Store cache

Step 1

Press:

Windows key + R

Step 2

Type:

```
wsreset.exe
```

Step 3

Press **Enter**.

Step 4

Wait for the command window to close automatically.

The Microsoft Store may open after the process is completed.

Completion

Microsoft Store cache reset command executed.

Process completed without an error.

Issues encountered

N/A

PART I: RUN WINDOWS DISK CLEANUP

Task 13: Open Disk Cleanup

Step 1

Press:

Windows key + R

Step 2

Type:

cleanmgr

Step 3

Press **Enter**.

Step 4

Select:

Local Disk (C:)

Step 5

Click **OK**.

Step 6

Select appropriate items such as:

Temporary Internet Files

Downloaded Program Files

Delivery Optimization Files

DirectX Shader Cache

Temporary files

Thumbnails

Windows error reports

Recycle Bin

Step 7

Click:

Clean up system files

Step 8

Select the **C:** drive again.

Step 9

Review the selected files carefully.

Step 10

Click **OK**, then select **Delete Files**.

Warning

Do not select the **Downloads** folder unless the user and ICT supervisor have confirmed that its contents can be deleted.

Completion

Disk Cleanup opened.

Appropriate cleanup categories selected.

Downloads folder was not deleted without approval.

Disk Cleanup completed successfully.

Issues encountered

N/A

PART J: CLEAR WINDOWS UPDATE DOWNLOAD CACHE

Task 14: Stop Windows Update services

Open PowerShell as Administrator.

Run:

```
Stop-Service -Name wuauserv -Force
```

Run:

```
Stop-Service -Name bits -Force
```

Completion

Windows Update service stopped.

Background Intelligent Transfer Service stopped.

Issues encountered

N/A

Task 15: Delete downloaded Windows Update cache files

Run:

```
Get-ChildItem -Path "C:\Windows\SoftwareDistribution\Download" -Force -  
ErrorAction SilentlyContinue | Remove-Item -Recurse -Force -ErrorAction  
SilentlyContinue
```

Completion

Windows Update download cache cleared.

Issues encountered

N/A

Task 16: Restart Windows Update services

Run:

```
Start-Service -Name bits
```

Run:

```
Start-Service -Name wuauserv
```

Completion

Background Intelligent Transfer Service restarted.

Windows Update service restarted.

Issues encountered

N/A

PART K: REPAIR WINDOWS COMPONENTS

Task 17: Run the Deployment Image Servicing and Management tool

Open PowerShell or Command Prompt as Administrator.

Run:

```
DISM.exe /Online /Cleanup-Image /RestoreHealth
```

Wait until the process reaches 100%.

Record the result

Operation completed successfully.

Corruption was detected and repaired.

The command completed with errors.

DISM result or error message:

Task 18: Run Windows System File Checker

Run:

```
sfc /scannow
```

Wait until the scan reaches 100%.

Record the result

Windows Resource Protection did not find integrity violations.

Corrupt files were found and repaired.

Corrupt files were found but some could not be repaired.

The scan did not complete.

SFC result or error message:

- Windows resource protection found corrupt files & successfully repaired them.

- For online repairs, details are included in the CBS log file
located: `winlog\logs\CBS\CBS.log`.

PART L: CHECK WINDOWS UPDATES

Task 19: Open Windows Update

Windows 11

Press:

Windows key + I

Go to:

Windows Update

Windows 10

Press:

Windows key + I

Go to:

Update & Security → Windows Update

Step 2

Click:

Check for updates

Step 3

Allow available updates to download and install.

Step 4

Do not interrupt the computer while updates are being installed.

Step 5

Record updates that require a restart.

Completion

Windows Update opened.

Update check completed.

Available security updates installed.

Available quality updates installed.

Driver updates reviewed by the ICT supervisor.

Restart requirement recorded.

Updates installed

Windows Upd Date

Issues encountered

N/A

PART N: RESTART THE COMPUTER

Task 23: Restart Windows

Before restarting:

All documents saved.

All programs closed.

User informed about the restart.

Windows Update completed or paused at a safe stage.

Restart using:

```
shutdown /r /t 0
```

Alternatively:

```
Get-CimInstance -ClassName Win32_ComputerSystem | Select-Object  
AutomaticManagedPagefile
```

Completion

Automatic virtual memory management confirmed.

Issues encountered

N/A

Task 26: Test basic computer functions

After restarting, test the following:

Computer starts normally.

Keyboard works.

Mouse works.

Internet connection works.

Office applications open.

Printer connection works.

Shared folders or network drives work.

Antivirus is active.

Windows Firewall is active.

No unexpected error messages appear.

Issues encountered

N/A

PART P: FINAL PRACTICAL CHECKLIST

Mark each completed activity.

Computer information recorded.

Initial free disk space recorded.

Start → Power → Restart

Completion

Computer restarted successfully.

Windows started normally after restart.

User successfully logged in.

Issues encountered

N/A

PART O: POST-CLEANUP CHECKS

Task 24: Confirm available disk space after cleanup

Run:

```
Get-PSDrive -Name C
```

Record the result

Free space before cleanup: 143 GB

Free space after cleanup: 144 GB

Space recovered: 1 GB

Completion

Disk space checked after cleanup.

Space recovered calculated and recorded.

Issues encountered

N/A

Task 25: Confirm the page-file setting

Run:

Restore point created.
User temporary files deleted.
Windows temporary files deleted.
Recycle Bin emptied.
DNS cache cleared.
Microsoft Store cache cleared.
Disk Cleanup completed.
Windows Update cache cleared.
Windows Update services restarted.
DISM repair command completed.
System File Checker completed.
Windows updates checked.
Approved updates installed.
Automatic virtual memory enabled.
Computer restarted.
Final free disk space recorded.
Basic computer functions tested.
Issues encountered documented.
Work verified by the ICT supervisor.

PART Q: ISSUES ENCOUNTERED DURING CLEANUP

No.	Task or Command	Issue Encountered	Error Message	Action Taken	Status
1					Resolved Pending
2					Resolved Pending
3					Resolved

4

Pending

Resolved

Pending

5

Resolved

Pending

6

Resolved

Pending

PART R: INTERN'S SUMMARY

1. What activities did you complete?

Used Disk Cleanup to remove temporary files.
Enable Storage Sense

2. How much disk space was recovered?

1 GB

3. What did you learn about virtual memory?

Proper configuration of virtual memory improves system stability and prevents crashes

4. Which command was most useful?

Cleanmgr.exe

5. What problems remain unresolved?

Certain temp files could not be deleted because they were running applications

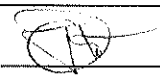
PART S: SUPERVISOR'S ASSESSMENT

Assessment Area	Maximum Marks	Marks Awarded
Observed safety and backup requirements	10	
Recorded computer	10	

information correctly	
Cleared temporary files and caches	20
Used commands correctly	15
Ran Windows repair tools	15
Checked and installed Windows updates	10
Configured virtual memory correctly	10
Documented issues and results	10
Total	100

Supervisor's comments

Intern's signature: _____ Date: _____

Supervisor's signature:  _____ Date: 24/7/26

